QUARK: A Quantum-Powered Index Advisor

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Abstract

Index configuration selection under storage constraints is a fundamental challenge in physical database design due to the exponential growth of the configuration space. The Index Advisor (IA) tools of contemporary database systems rely on greedy heuristics and often yield suboptimal recommendations. Recent quantum computing–based approaches overcome this limitation by exploiting quantum properties such as *superposition* and *entanglement*. In this demonstration, we present QUARK, a prototype quantum-powered index advisor. It leverages gate-based quantum computing and implements SQIA and OQIA, the *search-based* and *optimization-based* quantum algorithms proposed in [9], delivering higher-quality index configurations than existing solutions.

A novel technical contribution of QUARK is its support for *interpretability*. Specifically, it incrementally constructs, visualizes, and executes *partial* quantum circuits corresponding to the distinct logical steps of the quantum IA algorithms. At each step, intermediate quantum states are measured and interpreted using familiar DBMS concepts, providing insight into superposition-based configuration encoding, cost aggregation, and constraint enforcement.

Users interact with QUARK by specifying IA problem instances and exploring the alternative quantum approaches. They can execute the resulting quantum circuits on both noiseless simulators and real quantum hardware, including IBM's 156-qubit Heron processors. To quantify performance, the quantum-recommended index configurations are compared against a variety of baselines. Finally, a visual execution plan analysis is presented to understand the provenance of the observed performance differences. A sample video demonstrating QUARK's features is available at this link.

CCS Concepts

• **Information systems → Autonomous database administration**; • **Hardware → Quantum technologies**.

Keywords

Relational Index Advisor; Quantum Computing

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1 Introduction

Given an SQL query workload and a storage budget, the Index Advisor (IA) tool in an RDBMS recommends a suite of column indexes to improve workload execution performance while respecting the storage constraint. This index configuration selection problem is inherently combinatorial: the number of possible configurations grows exponentially with the number of candidate indexes, rendering exhaustive enumeration infeasible for realistic workloads. To cope with this complexity, commercial IA tools such as IBM's DB2 Index Advisor [14] and Microsoft's DTA [3] rely on greedy heuristic strategies. While these approaches scale well in practice, they often yield significantly sub-optimal recommendations, even losing up to 50% of achievable index benefits [9].

Given its significant impact on workload performance, IA design has been an active area of research for over three decades in both academia and industry. The literature includes sophisticated heuristic techniques and, more recently, ML and LLM-based approaches (e.g., [2, 3, 6, 10, 14]). Despite this rich body of work, the resulting performance improvements have remained incremental [17], underscoring the need for a fundamentally different paradigm.

Over the past few years, prototype quantum hardware has become a reality, and commercial-grade quantum computers (~ 2000 qubits and 1 Billion gates) are projected to be available by 2033 [5]. These advances have motivated the DBMS community to explore the potential of quantum computing for a variety of DBMS engine modules, including multi-query optimization [15], join-order optimization [11, 13], transaction scheduling [7], and index configuration [1, 9, 16]. In this demonstration, we present a quantum IA tool based on the hybrid classical-quantum **QIA** architecture introduced in [9], which represents the first effort to make index selection amenable to gate-based quantum computing.

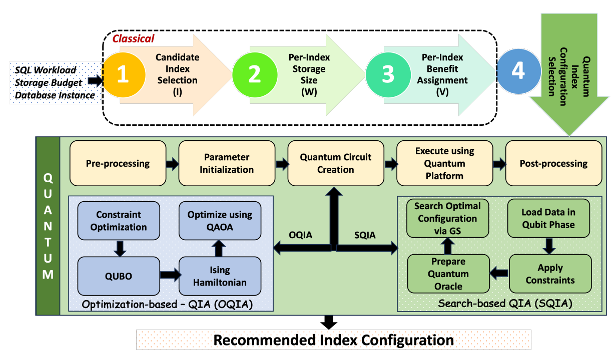

Figure 1: Quantum Index Advisor (QIA) Architecture [9]

Quantum Index Advisor (QIA) [9]

In contemporary systems, Index Advisors typically operate in four stages: (1) candidate index selection, (2) index storage size computation, (3) index benefit estimation, and (4) index configuration selection. QIA, whose architecture is shown in Figure 1, retains this conventional pipeline for the initial three stages. However, the final and computationally dominant step of index configuration selection is outsourced to a quantum platform.

QIA exploits the core principles of quantum computation. First, *quantum superposition* enables all index configurations to be represented simultaneously as computational basis states of a quantum system, making it feasible to explore the exponentially large configuration space in parallel. Second, *phase-based encoding* associates problem-specific quantities (e.g. aggregate storage costs, workload benefits, and feasibility under budget constraints) with individual configurations by applying controlled phase rotations.

Building on this foundation, QIA proposed two quantum algorithms for index configuration selection. The *Search-based Quantum Index Advisor* (SQIA) formulates the problem as a Grover-style quantum search [8] over the exponential configuration space. A problem-specific *quantum oracle* operates on a superposition of all index configurations, encoding storage size and workload benefit information and marking configurations that satisfy the storage constraint. Grover-style amplitude amplification is then used to bias measurement outcomes toward these qualifying configurations. This approach achieves a *quadratic* speedup over exhaustive search.

An alternative approach is the *Optimization-based Quantum Index Advisor* (OQIA). It formulates index configuration selection as a constrained optimization problem expressed as a Quadratic Unconstrained Binary Optimization (QUBO) instance and is solved using the Quantum Approximate Optimization Algorithm (QAOA) [4]. QAOA is a hybrid quantum-classical algorithm that iteratively refines a parameterized quantum circuit, using classical feedback to improve solution quality across iterations.

The QUARK System

In this demo, we introduce **QUARK** (*Quantum-powered Index Advisor for Relational Search Keys*), a prototype tool that fully implements the QIA framework, including both SQIA and OQIA, in an accessible and interpretable system.

Apart from producing high-quality index recommendations using a quantum computer, QUARK also features: (i) Seamless integration with commercial DBMS engines, and (ii) Visual interpretability of quantum execution. Achieving interpretability is difficult because quantum programs evolve through unitary operations within a closed system and do not naturally expose intermediate execution logs. To address this challenge, QUARK decomposes quantum execution into a sequence of semantically meaningful steps. For each step, the system constructs and executes a *prefix* quantum circuit that includes all quantum operations up to that point, followed by measurement. By progressively extending the circuit in this manner, QUARK enables users to observe the measured quantum states corresponding to successive steps of: (a) superposition-based configuration encoding, (b) aggregate storage size computation, (c) aggregate estimated benefit computation, and (d) storage constraint enforcement.

QUARK enables users to interactively configure IA problem instances, select quantum backends (noiseless simulators or real quantum hardware), and execute the resulting circuits. It compares the QUARK-recommended index configurations against several baselines, including the no-index case (*ground state*), commercial index advisors (*representing SOTA*), and, for tractable instances, the optimal configuration obtained via exhaustive enumeration (*upper bound*). These comparisons cover multiple dimensions, including estimated cost, execution time, and storage usage. Finally, a visual execution-plan analysis aids in understanding why the QUARK configurations deliver superior performance.

2 Demonstration Workflow

This section illustrates how users interact with the QUARK system to solve IA problem instances using a quantum computer.

2.1 Input IA Problem Instance

QUARK enables users to solve custom IA problem instances. Figure 2 illustrates the user interface, organized into three logical panels corresponding to the key inputs required by the advisor.

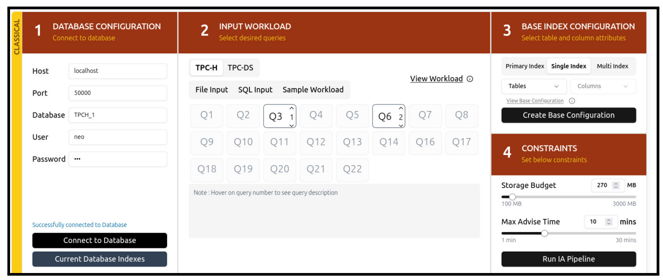

Figure 2: Input IA Problem Instance

The left panel establishes a live connection to a database instance and facilitates inspection of indexes currently deployed in the system. The central panel helps select the input workload. Users can choose queries from a benchmark suite (TPC-H or TPC-DS) and adjust their frequencies. Alternatively, they can directly provide custom SQL queries. Finally, the right panel optionally supports defining a base index configuration by selecting mandatory single and multi-column indexes that must be included in *all* recommended configurations. Users then specify a storage budget constraint using an interactive slider, along with a maximum advisory time limit.

2.2 Classical Stages of QUARK

QUARK executes the first three stages (candidate index selection, index storage size computation, and index benefit estimation) of the IA pipeline on a classical computing platform and extracts their outputs. Figure 3 visualizes these artifacts, which are subsequently passed as inputs to the quantum index selection stage, where either SQIA or OQIA can be invoked, as described next.

2.3 Index Selection with SQIA

QUARK constructs the SQIA oracle incrementally, following the progressive encoding strategy summarized in the Introduction. We illustrate this process using a simple IA instance with two candidate

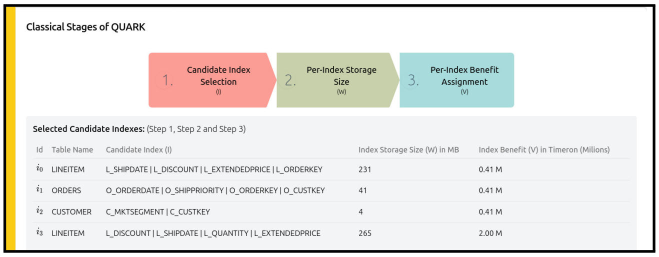

Figure 3: Classical Stages of QUARK

indexes. To facilitate visualization and explanation of the quantum circuit, we use normalized cost units. The two indexes have storage footprints of 1 and 4, and estimated workload benefits of 2 and 4, respectively, under a storage budget of 4. The corresponding quantum circuit is shown in Figure 4, where the oracle is decomposed into a sequence of six semantic steps, **S1** through **S6**. These steps correspond to: (S1) initialization and superposition of index configurations, (S2) computing configuration storage size, (S3) computing configuration workload benefit, (S4) constraint enforcement, (S5) converting the phase-encoded information into a measurable basis representation using the inverse quantum Fourier transform (`qft_dg`), and (S6) marking qualifying configurations.

Rather than treating the oracle as a monolithic black box, QUARK measures and visualizes the quantum state after each step. This provides insight into how the quantum state spans all index configurations, with each configuration implicitly associated with its aggregate storage size and workload benefit in superposition.

To interpret these states, each measured bitstring is decoded by partitioning its bits into four semantic components (from left to right): (i) a qualification indicator bit, (ii) bits encoding the aggregate workload benefit, (iii) bits encoding the aggregate storage size, and (iv) bits representing the index configuration itself. This structured mapping enables users to directly relate quantum states to concrete database concepts, making the behavior of the SQIA oracle transparent and interpretable despite its quantum execution.

2.4 Index Selection with OQIA

QUARK also supports OQIA, which formulates index configuration selection as a constrained optimization problem. The problem is transformed into a Quadratic Unconstrained Binary Optimization (QUBO) [12] representation and subsequently mapped to an equivalent Ising Hamiltonian for execution using the Quantum Approximate Optimization Algorithm [4]. QUARK visualizes each transformation stage, facilitating users to inspect the constructed objective function, the resulting QUBO terms, and the Hamiltonian coefficients prior to quantum execution. Details on transforming IA instances for QAOA are provided in Section 4 of [9].

2.5 Quantum Execution

QUARK supports execution on both noiseless quantum simulators and real IBM quantum hardware (at this time, IBM's 156-qubit Heron processors) hosted in the cloud. It exposes relevant quantum execution parameters, such as the number of *"shots"*, that is, the number of times a quantum circuit is executed. Higher shot counts improve result fidelity at the cost of increased quantum resource usage. The system transparently handles backend-specific steps, including circuit transpilation based on hardware characteristics. Upon completion, the measured quantum results are decoded to identify the index configuration recommended by QUARK.

2.6 Comparison with Baselines

QUARK enables direct comparison of index configurations recommended by the quantum approaches (SQIA/OQIA) against three baselines: (i) a no-index configuration, (ii) a SOTA configuration, as represented by IBM DB2, and (iii) the optimal configuration obtained via exhaustive enumeration for tractable IA instances. Figure 5 presents a quantitative comparison across multiple dimensions, including estimated execution cost and workload execution time benefit. When the optimal configuration is available, the interface reports the quality of each recommendation relative to this ideal, enabling users to assess both the sub-optimality of the SOTA recommendation and the proximity of the QUARK-recommended configuration to the optimal result. To further explain the observed performance differences, QUARK provides execution plan visualizations for workload queries.

3 Demonstration Scenario

The QUARK demonstration is organized into a sequence of interactive scenarios, spanning IA problem specification, quantum circuit construction and visualization, quantum execution, and comparative result analysis against baselines.

Scenario 1: IA Problem Configuration & Classical Stages Execution. Attendees specify an IA problem instance by defining an SQL query workload, an optional base index configuration, and storage and advisory time constraints. QUARK executes the first three stages of the classical IA pipeline and visualizes their outputs.

Scenario 2: Quantum Circuit Construction & Staged Execution. Attendees select either the SQIA or the OQIA approach and incrementally construct the quantum circuits as a sequence of semantically meaningful steps. Intermediate quantum states are visualized, allowing users to observe index configurations in superposition and how costs and constraints are encoded. This staged visualization provides transparency into the quantum execution and lowers the barrier to understanding quantum index selection for users lacking prior familiarity with quantum computing.

Scenario 3: Quantum Backend Execution. Attendees then execute the constructed circuits on a selected quantum backend, choosing between noiseless simulators and real quantum hardware. By varying execution-level parameters such as the number of shots, they can directly observe how backend choice and tuning decisions affect result stability and measurement outcomes.

Scenario 4: Configuration Comparison and Plan Analysis. Attendees are presented with quantitative comparisons of estimated execution cost, total time benefit, and storage consumption, allowing them to directly assess the quality of the quantum-recommended configurations. To explain the observed differences, the execution plans in the purely classical world and those obtained

Figure 4: Stepwise Construction of the SQIA Oracle Quantum Circuit Showing Logical Steps S1–S6

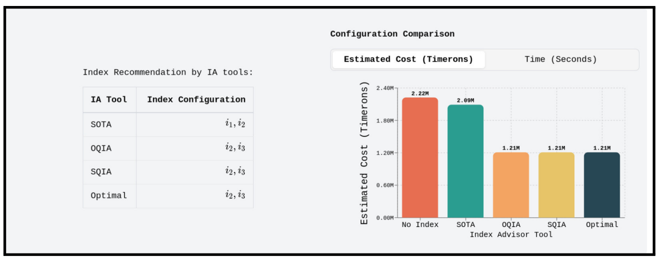

Figure 5: Comparison of Index Recommendation by IA Tools

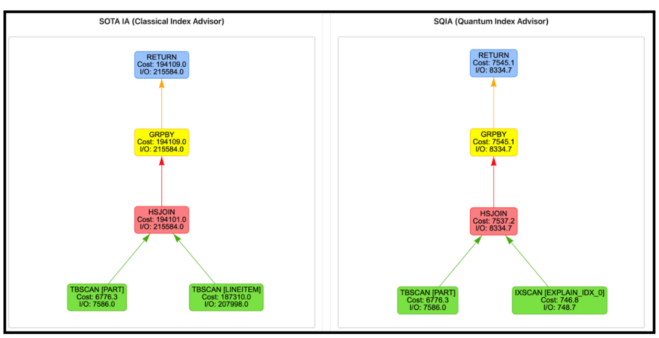

Figure 6: Index Utility through Execution Plan Comparison

with QUARK are shown side-by-side, enabling attendees to understand how index selection decisions propagate into observed improvements. For example, Figure 6 illustrates how a full table scan on `LINEITEM` is replaced by an index scan under the QUARK-recommended configuration, leading to a substantial reduction in estimated query cost from `194K` to `7.5K` timerons.

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